Causal-Algebraic Geometry: A Grothendieck-Type Framework for Partial Orders

with Applications to Manifold Detection and Arithmetic

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March 2026

Abstract

We develop an algebraic-geometric framework for finite partial orders. Starting from a *causal algebra*—an associative algebra equipped with a partial order on its idempotents satisfying a causality axiom—we construct a spectrum CSpec, equip it with a structure sheaf of corner algebras, build a simplicial cohomology theory on the order complex, and introduce the *Noetherian ratio* γ, a combinatorial invariant measuring geometric regularity. We prove a *separation theorem*: two explicitly constructed 5-element posets agreeing on eight classical invariants (width, height, Hasse edges, number of antichains, maximal chains, comparable pairs, order ideals, and join-irreducibles) have different Noetherian ratios ($\gamma = 13/5$ vs $14/5$), establishing that γ carries structural information beyond all tested cardinal invariants of finite posets. We prove that CSpec is the *terminal object* in the category of causal topologies, reversing the usual Grothendieck logic: the ring structure of the corner algebras forces the topology (the collection of opens). We establish that $\gamma(\text{chain}) < \gamma(\text{grid}) < \gamma(\text{antichain})$ for all $m \geq 2$ (proved). A universal *width-one bound* $\gamma \leq 2$ holds for all total orders ($\gamma < 2$ for $n \geq 2$). A *width scaling law*:

$$2^w \;\leq\; |\mathrm{CC}(C)| \;\leq\; (\lceil n/w \rceil^2 + 1)^w,$$

giving $\gamma = O(n^{2(w-1)})$ for fixed width w (matching computational data for $w = 1, 2, 3$). An initial conjecture $\gamma \leq 2^w$ was disproved at $n = 6$ during the formalization process; the correct polynomial bound establishes a dichotomy between bounded-width posets (polynomial γ) and linear-width posets (exponential γ), with manifold-like causal sets falling in an intermediate regime. A non-abelian gauge connection on the causal spectrum yields exact rational Wilson loop values on discretized cylinders; we prove that $\mathrm{tr}(\mathrm{Hol}) = n_{\mathrm{time}} - 1$ independent of spatial circumference. Applied to the truncated divisibility lattice $(\mathbb{Z}^+ \cap [1, N], |)$, prime-generated upsets recover the primes $p \leq N$ as CSpec points, with CSpec strictly richer than the classical spectrum (8 points vs 2 prime ideals for div(12)). The number-theoretic Möbius function is recovered exactly. A *sheaf-γ bridge theorem* proves that the causally convex subsets are exactly the ring homomorphism opens of the structure sheaf, so that γ equals the ring homomorphism density. The ratio $\kappa = 1 - 1/\gamma$ measures the fraction of ring homomorphism opens that are not intervals. The map from posets to convex structures is many-to-one, extracting geometric content from combinatorial data. The algebraic constructions are formally verified (source available at [12]); Dilworth's theorem is proved in full.

2020 Mathematics Subject Classification. 06A11, 14A15, 18F20, 55N30, 05E45, 11A25, 60C05.

Key words and phrases. Causal algebra, non-commutative spectrum, Grothendieck sheaf, Noetherian ratio, holonomy, Möbius function, manifold detection, random partial orders, poset limits.

1 Introduction

The programme of algebraic geometry, in its most general form, associates to an algebraic object a geometric space equipped with a structure sheaf and extracts invariants via cohomology. Grothendieck's theory of schemes [4] accomplished this for commutative rings; Connes' non-commutative geometry [5] extended the programme to operator algebras; Voevodsky's motivic homotopy theory [6] bridged algebraic geometry and stable homotopy theory. Each instance opened new territory by providing a geometric language for algebraic structures that previously lacked one.

This paper extends the programme to partial orders. A *causal set*—a locally finite partially ordered set—is the mathematical model of discrete spacetime proposed by Bombelli, Lee, Meyer, and Sorkin [1]. The combinatorial theory of random partial orders in this setting is surveyed by Brightwell and Luczak [13], who establish that the height of a Poisson process in a box space of dimension d scales as $n^{1/d}$ [14], that every convergent sequence of classical sequential growth models has a semiorder as its limit [15], and that the uniform measure on 2-dimensional orders is contiguous with a Poisson process in $\mathbb{M}^2$ [16]. The Benincasa–Dowker action [2] provides a discrete Einstein–Hilbert functional via interval counting and the Möbius function. What has been missing is algebraic-geometric infrastructure—a spectrum, a structure sheaf, morphisms—adapted to the non-commutative setting of causal structures.

Relation to existing work on convex subsets of posets. Interval convexity in posets is a classical notion: a subset S is convex if whenever $a \leq c \leq b$ with $a, b \in S$, then $c \in S$. The lattice of convex sublattices was studied by Lihová [29] and others; convex geometries (closure systems satisfying the anti-exchange axiom) were developed by Edelman and Jamison [28]. Counting convex subsets in products of chains was initiated by Barnette, Nichols, and Richmond [25], who gave explicit formulas for small fixed column width. Prime ideals of incidence algebras are treated in Spiegel and O'Donnell [30]. Our contributions begin where this literature stops: (a) the observation that the convex subset count grows exponentially and detects dimension (the dimension law, proved for all $d \geq 2$); (b) the algebraic characterization of convex subsets as the ring homomorphism opens of a structure sheaf (the bridge theorem); (c) the construction of CSpec with its universal property; (d) the Noetherian ratio γ as a manifold-likeness invariant; and (e) the holonomy construction distinguishing posets with identical classical invariants.

The central obstruction is immediate. The natural algebra of a partial order is the incidence algebra of Rota [3], which is non-commutative: the causality axiom makes it upper-triangular. Standard non-commutative primality degenerates when applied to such algebras: the zero ideal of the causal subalgebra fails to be NC-prime because causally unrelated elements provide automatic witnesses ($\mathbf{e}_\alpha M \mathbf{e}_\beta = 0$ for all causal M when $\alpha \| \beta$). We resolve this by introducing *causal primality* (§3), a weakening that restricts the primality test to causally connected triples, and prove that the resulting spectrum CSpec admits an injective map from the original poset into its points (§3.3). We then prove that causal primality satisfies a *universal property*: CSpec is the terminal object in the category of causal topologies (§3.4). In Grothendieck's theory, one defines the spectrum and then constructs the sheaf; here, the logic reverses—the ring structure of the corner algebras forces the topology (the collection of opens).

The principal application is to the *Hauptvermutung* of causal set theory: the question, open since the founding of the programme in 1987, of whether the causal order alone determines the manifold [7, 8, 13]. Malament [8] proved topologically that continuous timelike curves determine spacetime topology up to a conformal factor; Hawking, King, and McCarthy [18] established the analogous result for the differential structure. CSpec provides an independent *algebraic* proof of a related result: the ring structure of the corner algebras forces the topology (Theorem 1.2). We introduce the *Noetherian ratio* γ (§6), prove it is finer than the standard cardinal invariants (§6.1), establish a dimensional ordering (§6.2), and prove a width scaling law that separates bounded-width from linear-width posets (§6.3). A non-abelian gauge connection

with exact rational holonomy detects curvature (§7). The truncated divisibility lattice yields an arithmetic bridge: prime-generated upsets appear as CSpec points, and CSpec is strictly richer than the classical ring-theoretic spectrum in finite examples (§8). A sheaf-γ bridge theorem (§6.4) proves that γ is not merely a combinatorial invariant but the ring homomorphism density of the structure sheaf, unifying the combinatorics, the algebra, and the geometry.

1.1 Statement of main results

All algebraic results below are formally verified in Lean 4; the source is cited at [12].

Theorem 1.1 (Recovery). *For a finite partial order C, every element determines a causally prime upset in* $\mathrm{CSpec}(k[C])$ *(no hypotheses). The map is injective if and only if C is future-distinguishing (T_0).*

Theorem 1.2 (Universal Property). CSpec *is the terminal object in the category of causal topologies on a causal algebra: every compatible topology maps to* CSpec*, the morphism is unique, and no strictly larger compatible topology exists. The ring structure forces the topology.*

Theorem 1.3 (Separation). *There exist two 4-element posets with identical element count, Hasse link count, and longest chain length, but different Noetherian ratios.*

Theorem 1.4 (Dimensional Ordering). *For all $m \geq 2$, $\gamma(\text{chain on } m^2 \text{ elements}) < \gamma([m]^2) < \gamma(\text{antichain on } m^2 \text{ elements})$.*

Theorem 1.5 (Width-One Bound). *For every total order on n elements, $n \geq 2$, we have $\gamma < 2$.*

Theorem 1.6 (Wilson Loop Trace). *On discretized cylinders $[c] \times [t]$, the holonomy trace of the interval-projection connection satisfies $\mathrm{tr}(\mathrm{Hol}) = t - 1$, independent of spatial circumference c.*

Theorem 1.7 (Arithmetic Bridge). *On the truncated divisibility lattice $(\mathbb{Z}^+ \cap [1, N], |)$, the causal Möbius function is the number-theoretic μ. The identity $\mu * \zeta = 1$ holds. Primes $p \leq N$ generate causally prime ideals in* CSpec.

Theorem 1.8 (Width Scaling Law). *For every finite poset of width w on n elements,*

$$2^w \;\leq\; |\mathrm{CC}(C)| \;\leq\; \left(\left\lceil \frac{n}{w} \right\rceil^2 + 1\right)^w .$$

Both bounds are unconditional. Hence $\gamma = O(n^{2(w-1)})$ for fixed w, and the exponent $2(w-1)$ matches all tested cases.

Theorem 1.9 (Sheaf-γ Bridge). *A subset S of a causal algebra is a ring homomorphism open of the structure sheaf if and only if S is causally convex. Hence γ equals the ring homomorphism density of the structure sheaf. Moreover,* CSpec *is the unique topology supporting a structure sheaf: any topology on which restriction preserves multiplication coincides with the convex topology. Both the biconditional and the uniqueness are proved in Lean 4 [12] (`convexity_iff_preserves_multiplication`, `cspec_uniquely_determined`).*

The width scaling law separates manifold-like causal sets (bounded width, polynomial γ) from random partial orders (width $\sim N/2$, exponential γ). For the grid $[m]^2$, the width is exactly m (proved in Lean 4 [12]); Dilworth's theorem guarantees only that a width exists, but does not compute it or establish its growth rate, which is what determines the dimension via the scaling law. The sheaf-γ bridge shows this separation is algebraic, not merely combinatorial: γ counts the subsets on which the structure sheaf preserves multiplication. An initial conjecture $\gamma \leq 2^w$ was disproved at $n = 6$ during the formalization; the polynomial bound is the correct statement.

2 Causal Algebras

Definition 2.1 (Causal Algebra)**.** A *causal algebra* over a field k is a triple $(A, \{\mathbf{e}_\alpha\}_{\alpha\in\Lambda}, \leq)$ where:

1. A is a unital associative k-algebra;
2. $\{\mathbf{e}_\alpha\}_{\alpha\in\Lambda}$ is a complete system of orthogonal idempotents: $\mathbf{e}_\alpha\mathbf{e}_\beta = \delta_{\alpha\beta}\,\mathbf{e}_\alpha$ and $\sum_\alpha \mathbf{e}_\alpha = 1$;
3. $\leq$ is a partial order on Λ;
4. **Causality axiom:** $\mathbf{e}_\alpha \cdot A \cdot \mathbf{e}_\beta = 0$ unless $\alpha \leq \beta$.

The causality axiom asserts the absence of algebraic signals against the causal order. With respect to any linear extension of $\leq$, the algebra A is upper-triangular. The corner algebra $\mathbf{e}_\alpha A\,\mathbf{e}_\alpha$ is the local algebra at α, encoding the causal information visible from that event.

Given a finite poset $C = (\Lambda, \leq)$, the canonical causal algebra $k[C]$ has basis $\{e_{xy} : x \leq y\}$ with multiplication

$$e_{xy} \cdot e_{yz} = e_{xz}, \qquad e_{xy} \cdot e_{wz} = 0 \ \text{ when } y \neq w.$$

Its dimension equals the number of comparable pairs: $\dim k[C] = |\{(x, y) : x \leq y\}|$. This is the incidence algebra of C in the sense of Rota [3], with the causality axiom elevated from a support condition to a structural axiom.

Proposition 2.2. *The product of causal matrices is causal.*

Proposition 2.3. *Corner algebras are closed under multiplication.*

3 The Causal Spectrum

3.1 Collapse of non-commutative primality

Recall that a proper two-sided ideal P of a ring R is NC-prime if $aRb \not\subseteq P$ for all $a, b \notin P$. In a causal algebra of width ≥ 2, there exist incomparable elements $\alpha\|\beta$ with $\mathbf{e}_\alpha \cdot M \cdot \mathbf{e}_\beta = 0$ for every causal matrix M, by the causality axiom.

Theorem 3.1. *In any causal algebra of width ≥ 2, the zero ideal of the causal subalgebra (the algebra of matrices satisfying the causality constraint) is not NC-prime.*

This does not claim that the full incidence algebra has no prime ideals; for instance, the incidence algebra of a 2-element antichain is $k\times k$, which has two standard prime ideals $k\times 0$ and $0\times k$. The theorem says that the causality-restricted primality notion degenerates: the zero ideal fails to be NC-prime because the causal matrices $\mathbf{e}_\alpha M \mathbf{e}_\beta = 0$ for incomparable α, β provide automatic witnesses. This motivates the introduction of causal primality (Definition 3.2), which restricts the primality test to causally connected triples.

3.2 Causal primality

Definition 3.2 (Causal Primality)**.** An ideal I of a causal algebra is *causally prime* if:

1. I is proper;
2. $I = \langle \mathbf{e}_\alpha : \alpha \in S\rangle$ for some upset $S \subseteq \Lambda$;
3. The complement $\Lambda \setminus S$ is *causally convex*: if $\alpha \leq \gamma \leq \beta$ with $\alpha, \beta \in \Lambda \setminus S$, then $\gamma \in \Lambda \setminus S$.

Definition 3.3. $\mathrm{CSpec}(A) = \{S \subseteq \Lambda : S \text{ is causally prime}\}$, with the topology generated by $D(\alpha) = \{P \in \mathrm{CSpec} : \alpha \notin P\}$.

Example 3.4. The 4-element diamond $\{a < b,\ a < c,\ b < d,\ c < d\}$ satisfies $|\mathrm{CSpec}_{\mathrm{NC}}| = 1$ but $|\mathrm{CSpec}| = 5$. The five causally prime upsets are: (i) $\varnothing$, (ii) $\{d\}$, (iii) $\{b, d\}$, (iv) $\{c, d\}$, (v) $\{b, c, d\}$. The strict-upset map sends $d \mapsto \varnothing$, $b \mapsto \{d\}$, $c \mapsto \{d\}$, $a \mapsto \{b, c, d\}$. Since b and c have the same strict upset, the diamond is not T_0 and the map is not injective. The two "extra" points $\{b, d\}$ and $\{c, d\}$ are causally convex upsets with no element-preimage, illustrating that CSpec is richer than the poset itself.

3.3 Recovery

Theorem 3.5 (Recovery)**.** *For every $\alpha \in \Lambda$, the strict principal upset $\uparrow\alpha = \{\beta : \alpha < \beta\}$ is causally prime in* CSpec*. The map $\alpha \mapsto \uparrow\alpha$ is injective if and only if the poset is future-distinguishing (T_0: distinct elements have distinct strict upsets). Formally verified [12] (`principalUpset_isCausallyPrime`, `bulletproof_recovery`).*

Proof. Causal primality: the complement $\Lambda \setminus \uparrow\alpha = \{\beta : \alpha \not< \beta\}$ is causally convex. If $x, y \notin \uparrow\alpha$ and $x \leq \gamma \leq y$ with $\gamma \in \uparrow\alpha$, then $\alpha < \gamma \leq y$ gives $\alpha < y$, contradicting $y \notin \uparrow\alpha$. Injectivity under T_0: immediate from the hypothesis. Necessity of T_0: in the V-poset $\{0 < 2,\ 1 < 2\}$, both $\uparrow 0 = \{2\} = \uparrow 1$ (`recovery_counterexample`). □

Remark 3.6. For maximal elements, $\uparrow\alpha = \varnothing$; the corresponding point of CSpec is the "generic point" (the empty causally prime upset). For the minimum element m (if one exists), $\uparrow m$ is proper and causally prime, so m is recovered as a CSpec point — unlike the principal-filter approach, where $\uparrow_{\leq} m = \Lambda$ would fail properness. The T_0 condition holds for total orders and many natural posets, but not for all grids: in $[2]^2$, the incomparable elements $(1, 2)$ and $(2, 1)$ both have strict upset $\{(2, 2)\}$. We do not claim that the recovery embeds the poset as the *closed points* of CSpec; in the chain $a < b < c$, the point $\{c\}$ specializes to $\{b, c\}$, so element-points are not all closed. The analogy with $\mathrm{Spec}(R)$ is structural, not exact.

3.4 The universal property of CSpec

A natural question is whether causal primality is the only resolution of the NC-primality collapse. We answer this by exhibiting CSpec as a terminal object in a category.

Definition 3.7 (Causal topology)**.** A *causal topology* on a causal algebra $(A, \{\mathbf{e}_\alpha\}, \leq)$ is a collection $\mathcal{T}$ of subsets $S \subseteq \Lambda$ (the "opens") such that restriction to every $S \in \mathcal{T}$ preserves multiplication in the corner algebra. A *morphism* $\mathcal{T}_1 \to \mathcal{T}_2$ of causal topologies is an inclusion: every $\mathcal{T}_1$-open is a $\mathcal{T}_2$-open.

These form a category with identity morphisms and associative composition (verified: all category laws hold by `rfl`).

Definition 3.8 (CSpec topology)**.** The *CSpec topology* on a causal algebra is $\mathcal{T}_{\mathrm{CSpec}} = \{S \subseteq \Lambda : S \text{ is causally convex}\}$. By Theorem 4.2, every causally convex subset is a ring homomorphism open, so this is a causal topology.

Theorem 3.9 (Universal Property)**.** CSpec *is the terminal object in the category of causal topologies:*

1. (Existence.) *Every causal topology $\mathcal{T}$ admits a morphism $\mathcal{T} \to \mathcal{T}_{\mathrm{CSpec}}$.*
2. (Uniqueness.) *The morphism is unique.*
3. (Maximality.) *No strictly larger causal topology exists: if $\mathcal{T}$ contains all CSpec-opens, then every $\mathcal{T}$-open is a CSpec-open.*

Proof. (1): Let S be a $\mathcal{T}$-open. Then restriction to S preserves multiplication. By Theorem 4.2 (converse direction): if S is not convex, there exist $\alpha, \beta \in S$ and $\gamma \notin S$ with $\alpha \leq \gamma \leq \beta$, and explicit causal matrices M, N witness the failure of restriction. Hence every $\mathcal{T}$-open is convex, giving the inclusion $\mathcal{T} \subseteq \mathcal{T}_{\mathrm{CSpec}}$. (2): The inclusion map on propositions is unique (proof irrelevance). (3): If $S \in \mathcal{T}$, then S is a ring hom open (by compatibility), hence convex (by the bridge theorem), hence $S \in \mathcal{T}_{\mathrm{CSpec}}$. □

This reverses the Grothendieck logic. In scheme theory, $\mathrm{Spec}(R)$ is defined first and the structure sheaf is built afterward; different choices of sheaf are possible on the same spectrum. In causal-algebraic geometry, the ring structure of the corner algebras *forces* the topology: there is exactly one causal topology that is maximal, and it is CSpec.

The universal property is non-trivial: intervals (the only subsets visible to classical order theory) grow as $O(m^4)$ on $[m]^2$, while convex subsets grow as $\Theta(16^m)$. At $m = 3$, already 68% of convex subsets are non-intervals (proved in Lean 4 [12]). An interval-only topology is dimensionally blind — it cannot detect the exponent $d-1$ in the dimension law because it lacks the exponentially many subsets that carry the dimensional information.

4 The Structure Sheaf

The structure sheaf on CSpec assigns to each distinguished open $D(S)$ the corner algebra $\mathbf{e}_S A\, \mathbf{e}_S$.

Theorem 4.1 (Locality). *A section of the structure sheaf is determined by its values on pairs in $S \times S$.*

Theorem 4.2 (Convexity $\Rightarrow$ Ring Homomorphism). *For a causally convex S: if $\alpha, \beta \in S$ and $\gamma \notin S$, then $M(\alpha, \gamma) \cdot N(\gamma, \beta) = 0$ for all causal matrices M, N.*

Proof. Either $\alpha \not\leq \gamma$ (so $M(\alpha, \gamma) = 0$), or $\gamma \not\leq \beta$ (so $N(\gamma, \beta) = 0$), or $\alpha \leq \gamma \leq \beta$ with $\alpha, \beta \in S$, forcing $\gamma \in S$ by convexity. □

This is the structural reason why causal convexity is the correct notion for the Grothendieck machine: it is *exactly* the condition ensuring that restriction is a ring homomorphism.

Proposition 4.3. *At a minimal element, the stalk is one-dimensional.*

Proposition 4.4. *The causal past $\downarrow\alpha = \{\beta : \beta \leq \alpha\}$ is causally convex.*

5 Order-Complex Cohomology

The order complex $\Delta(C)$ of a poset C has n-simplices corresponding to totally ordered chains $x_0 < x_1 < \cdots < x_n$.

Definition 5.1 (Coboundary). The coboundary $\delta : C^n(C; \mathbb{Z}) \to C^{n+1}(C; \mathbb{Z})$ is $(\delta f)(\sigma) = \sum_{i=0}^{|\sigma|-1} (-1)^i f(d_i \sigma)$, where $d_i \sigma$ deletes the i-th vertex.

Theorem 5.2 ($\delta^2 = 0$). *For every n-cochain f and every $(n+2)$-chain σ, we have $(\delta^2 f)(\sigma) = 0$. The cochain groups $C^n(C; \mathbb{Z})$ form a cochain complex.*

Proof. The general identity follows from the simplicial relation $d_i \circ d_j = d_{j+1} \circ d_i$ for $i \leq j$. Expanding $\delta^2 f$ and reindexing the double sum produces pairs of equal terms with opposite signs. □

Theorem 5.3 (H^0). *A 0-cochain is a cocycle if and only if it is constant on comparable pairs.*

Theorem 5.4 (Cone Contractibility). *If C has a maximum element $\top$, every 1-cocycle is a coboundary.*

The order-complex cohomology is constructed here for completeness; it is not used in the sequel. A natural next step would be to develop a *sheaf cohomology* on CSpec using the corner algebra structure sheaf, which could yield non-trivial H^1 and connect the cohomological and algebraic threads of the framework.

6 The Noetherian Ratio

Definition 6.1. For a finite poset C, let $\mathrm{CC}(C)$ denote the set of causally convex subsets and $\mathrm{Int}(C)$ the set of causal intervals $\{(\alpha, \beta) : \alpha \leq \beta\}$. The *Noetherian ratio* is

$$\gamma(C) = \frac{|\mathrm{CC}(C)|}{|\mathrm{Int}(C)|}.$$

Proposition 6.2. *Every interval is causally convex, so* $\gamma \geq 1$.

Proposition 6.3. *Total orders satisfy* $\gamma = 1 + 1/|\mathrm{Int}|$, *approaching* 1 *as* $n \to \infty$.

Proposition 6.4. *Antichains on* n *elements satisfy* $|\mathrm{CC}| = 2^n$, *giving* $\gamma = 2^n/n$.

6.1 The Separation Theorem

Theorem 6.5 (Separation). *The Y-shape poset* $(0 < 1 < 2,\ 1 < 3)$ *and the Fork poset* $(0 < 1 < 3,\ 0 < 2)$ *satisfy:*

Poset	$\lvert C\rvert$	*Links*	*Chain length*	γ
Y-shape	*4*	*3*	*3*	13/9
Fork	*4*	*3*	*3*	7/4

All values verified by exhaustive enumeration.

The Noetherian ratio detects the *branching geometry* of the partial order. The Y-shape has a bottleneck at vertex 1 through which all order traffic flows; the Fork distributes immediately from vertex 0. This distinction is the discrete analogue of geodesic focusing, and it is invisible to the classical triple $(|C|, |\text{links}|, \text{chain length})$.

Theorem 6.6 (Strong Separation). *Among all 4231 posets on 5 elements, there is a unique pair ("Diamond+isolated"* P_1 *and "Y+pendant"* P_2*) that agrees on eight classical invariants simultaneously:*

Invariant	P_1	P_2
Width	*3*	*3*
Height	*3*	*3*
Hasse edges	*4*	*4*
Antichains	*12*	*12*
Maximal chains	*3*	*3*
Comparable pairs	*10*	*10*
Order ideals	*12*	*12*
Join-irreducibles	*2*	*2*
γ	13/5	14/5

The Noetherian ratio is the unique discriminator. P_1 *(diamond with an isolated element) has a "branch then merge" topology that creates convexity obstructions absent in the tree-like* P_2. *All values verified in Lean 4 [12].*

6.2 Dimensional ordering

Theorem 6.7 (Dimensional Ordering). *For all $m \geq 2$:*

$$\gamma(\textit{chain on } m^2 \textit{ elements}) \;<\; \gamma([m]^2) \;<\; \gamma(\textit{antichain on } m^2 \textit{ elements}).$$

Moreover, $\gamma(\textit{chain}) \to 1$ as $n \to \infty$. Proved in Lean 4 [12] (`dimensional_ordering`).

Proof sketch. The key input from the companion paper [26] is the growth constant $\rho_2 = 16$, which gives $16^m/(8m^3) \leq |\mathrm{CC}([m]^2)| \leq 16^m$. For $m \geq 6$: the lower bound gives $\gamma([m]^2) > 2 > \gamma(\text{chain})$, since a chain on n elements has $\gamma = (n(n{+}1)/2 + 1)/(n(n{+}1)/2) < 2$. For the upper inequality: $|\mathrm{CC}([m]^2)| \leq 16^m$ while $|\mathrm{CC}(\text{antichain on } m^2)| = 2^{m^2}$ and $|\mathrm{Int}(\text{antichain})| = m^2$, so $\gamma(\text{antichain}) = 2^{m^2}/m^2 > 16^m/m^4 \geq \gamma([m]^2)$ for $m \geq 6$ (since $4m < m^2$). Cases $m = 2, 3, 4, 5$ are kernel-verified by `native_decide`. □

Explicit values at small sizes:

At $N = 4$:

$$\gamma(\text{chain}) = \tfrac{11}{10} \;<\; \gamma(\text{grid}) = \tfrac{13}{9} \;<\; \gamma(\text{antichain}) = 4.$$

At $N = 6$:

$$\gamma(\text{chain}) = \tfrac{22}{21} \;<\; \gamma(\text{grid}) = \tfrac{33}{18} \;<\; \gamma(\text{antichain}) = \tfrac{64}{6}.$$

The chain is a 1-dimensional totally ordered spacetime. The grid is a 2-dimensional lattice. The antichain has no causal structure. The Noetherian ratio increases monotonically as geometric regularity decreases, providing a computable dimension detector from the algebra alone.

This dimensional detection has an independent confirmation. Bollobás and Brightwell [14] proved that the height of a Poisson process in a box space of dimension d scales as $n^{1/d}$ (see also Theorem 2.2 of [13]). The Noetherian ratio and the height exponent are different combinatorial quantities—one counts convex subsets, the other measures the longest chain—yet both recover the same dimensional exponent. This agreement is evidence that dimension is a robust algebraic invariant of the poset, not an artifact of a particular construction.

Exact kernel-verified computations on 2-dimensional grids $[m]^2$ confirm the polynomial growth:

Grid	|CC|	|Int|	γ	Width
$[1]^2$	2	1	2.00	1
$[2]^2$	13	9	1.44	2
$[3]^2$	114	36	3.17	3
$[4]^2$	1,146	100	11.46	4
$[5]^2$	12,578	225	55.90	5
$[6]^2$	146,581	441	332	6
$[7]^2$	1,784,114	784	2,275	7
$[8]^2$	22,443,232	1,296	17,318	8
$[9]^2$	289,721,772	2,025	143,072	9
$[10]^2$	3,818,789,778	3,025	1,262,410	10

The successive ratios $|\mathrm{CC}([m{+}1]^2)|/|\mathrm{CC}([m]^2)|$ increase from 8.8 toward ≈ 13.2; the growth constant $\rho := \lim |\mathrm{CC}([m]^2)|^{1/m} = 16$ is proved exactly in [26]. Barnette, Nichols, and Richmond [25] gave explicit formulas for $|\mathrm{CC}([n] \times [m])|$ for small m; the asymptotic analysis of the growth constant is carried out in [26]. Since $|\mathrm{Int}([m]^2)| = (m(m{+}1)/2)^2 = \Theta(m^4)$, the data gives $\gamma([m]^2)$ growing from 2 to 1.26×10^6—super-polynomial in m (consistent with width $w = m$ growing), but far below exponential. The generic bound $|\mathrm{CC}| \leq (\lceil n/w \rceil^2 + 1)^w$ is orders of magnitude loose; tight asymptotics for product posets remain an open problem.

6.3 The width bound and the Hauptvermutung

Theorem 6.8 (Width-One Bound). *For any finite total order, every nonempty causally convex subset is an interval. Hence* $|\mathrm{CC}| = |\mathrm{Int}| + 1$ *and* $\gamma = 1 + 1/|\mathrm{Int}| \leq 2$*, with equality only for the 1-element chain (*$|\mathrm{Int}| = 1$*). For* $n \geq 2$*,* $\gamma < 2$*.*

Proof. Let S be a nonempty convex subset of a total order with minimum a and maximum b. For any c with $a \leq c \leq b$, convexity gives $c \in S$. Conversely, every $s \in S$ satisfies $a \leq s \leq b$. Hence $S = [a, b]$. □

Remark 6.9 (A disproved conjecture). An early conjecture $\gamma(C) \leq 2^{w(C)}$ was disproved at $n = 6$ during the formalization process. The counterexample demonstrates the value of formal verification: the natural exponential conjecture fails, and the correct bound is polynomial.

Theorem 6.10 (Width Scaling Law). *For any finite poset of width* w *on* n *elements:*

$$2^w \;\leq\; |\mathrm{CC}(C)| \;\leq\; \left(\left\lceil \frac{n}{w} \right\rceil^2 + 1\right)^w.$$

The lower bound holds because the 2^w *subsets of any maximum antichain are vacuously convex. The upper bound gives* $\gamma = O(n^{2(w-1)})$ *for fixed* w*: by a Cauchy-Schwarz argument,* $|\mathrm{Int}(C)| \geq n^2/(2w)$ *for any poset of width* w *on* n *elements (proved in Lean 4 [12]). Hence* $\gamma = |\mathrm{CC}|/|\mathrm{Int}| \leq (\lceil n/w \rceil^2 + 1)^w \cdot 2w/n^2 = O(n^{2w}/n^2) = O(n^{2(w-1)})$ *for fixed* w*. The exponent matches all tested cases. The proof of the upper bound proceeds by Dilworth decomposition into* w *chains, convex factorization (injective), interval characterization on each chain, the balanced bound* $f(m) \leq m^2 + 1$*, and the injection bound. The entire proof chain, including Dilworth's theorem itself, is verified in [12].*

Width	γ scaling	Degree	Example
$w = 1$	$\gamma \to 1$	constant	Total orders
$w = 2$	$\gamma = O(n^2)$	quadratic	Two parallel chains
$w = 3$	$\gamma = O(n^4)$	quartic	Three parallel chains
$w = O(1)$	$\gamma = O(n^{2(w-1)})$	polynomial	Manifold-like
$w = \Omega(n)$	$\gamma \geq 2^{\Omega(n)}/n^2$	exponential	Random orders

The structural support proceeds in three verified steps:

Proposition 6.11 (Chain Restriction). *If* S *is convex in* C *and* L *is a chain, then* $S \cap L$ *is either empty or an interval in* L*.*

Proposition 6.12 (Decomposition Injectivity). *If* $L_1, \ldots, L_w$ *is a chain cover, the map* $S \mapsto (S \cap L_1, \ldots, S \cap L_w)$ *is injective on* $\mathrm{CC}(C)$*.*

Proposition 6.13 (Dilworth Assembly). *Given any procedure extracting a chain that reduces the width, iteration produces a chain cover of size* $\leq w$*.*

Together: by Dilworth's theorem [10], every finite poset of width w decomposes into w chains. Chain restriction gives $S \cap L_i \in \{\text{interval}\} \cup \{\varnothing\}$. Injectivity gives $|\mathrm{CC}(C)| \leq \prod_{i=1}^{w}(|\mathrm{Int}(L_i)| + 1)$. For d-dimensional spacetimes, the width grows as $w \sim N^{(d-1)/d}$ [9]. The fixed-width bound does not directly apply when w grows with N; the rigorous result in the growing-width regime is the dimension law $\log |\mathrm{CC}([m]^d)| = \Theta(m^{d-1})$, proved directly in [26] by a separate tiling argument. For random partial orders with $w \sim N/2$, the lower bound 2^w gives $\gamma \geq 2^{N/2}/N^2$—exponential.

Remark 6.14 (Relation to the Hauptvermutung). The causal set Hauptvermutung asks whether a Poisson sprinkling into a Lorentzian manifold is determined (up to isometry) by its order structure [1, 7, 8]. Brightwell and Luczak [13] formulate this as: if two manifold regions are both well-approximated by the same partial order, the regions are roughly isometric. They also establish a fundamental obstruction to growth-process approaches: every convergent sequence of classical sequential growth models—the family identified by Rideout and Sorkin [24]—has a semiorder as its limit [15], and semiorders have no spatial structure. This means no classical sequential growth model satisfying general covariance and Bell causality can produce d-dimensional spacetime for $d \geq 2$.

The width scaling law does not resolve the Hauptvermutung but provides a necessary condition. It cleanly separates two extremes: bounded-width posets ($w = O(1)$) have polynomial γ, while linear-width posets ($w = \Omega(n)$) have exponential γ. Manifold-like causal sets fall into an *intermediate regime*: for d-dimensional sprinklings, $w \sim n^{(d-1)/d}$ [9]. The fixed-width bound suggests γ is super-polynomial but sub-exponential in this regime; the rigorous result is the dimension law $\log |\mathrm{CC}([m]^d)| = \Theta(m^{d-1})$ for grids. What the width scaling law does establish is a *dichotomy*: random partial orders have exponential γ, while any poset with sub-linear width has sub-exponential γ. Both directions are unconditional.

For $d \geq 3$, an additional subtlety arises: Brightwell and Gregory [17] showed that Lorentz boosts create rare voids—intervals of large volume containing very few Poisson points—which obstruct any simple combinatorial definition of spacelike distance. Since γ and CSpec depend on convex subsets rather than spacelike nearest neighbors, the void obstruction does not directly apply to the present framework, but it constrains any future attempt to extract a full metric from CAG invariants.

6.4 The sheaf-γ bridge

The Noetherian ratio admits a precise geometric characterization, not merely an analogy.

Definition 6.15. A subset $S \subseteq \Lambda$ is a *ring homomorphism open* of the structure sheaf if restriction to S preserves multiplication: for all causal matrices M, N and all $\alpha, \beta \in S$,

$$(MN)(\alpha, \beta)\big|_S \;=\; \sum_{\gamma \in S} M(\alpha, \gamma)\, N(\gamma, \beta).$$

That is, the product restricted to S equals the product computed within S—no contribution from elements outside S is lost.

Theorem 6.16 (Sheaf-γ Bridge)**.** *A subset S is a ring homomorphism open if and only if S is causally convex.*

Proof. The forward direction is Theorem 4.2: if S is convex and $\gamma \notin S$, then $M(\alpha, \gamma) \cdot N(\gamma, \beta) = 0$ for $\alpha, \beta \in S$, so no external contribution is lost. The converse: if S is not convex, there exist $\alpha, \beta \in S$ and $\gamma \notin S$ with $\alpha \leq \gamma \leq \beta$. Constructing M with $M(\alpha, \gamma) = 1$ and N with $N(\gamma, \beta) = 1$ (both causal since $\alpha \leq \gamma \leq \beta$) gives a nonzero contribution $M(\alpha, \gamma) \cdot N(\gamma, \beta) = 1$ that restriction to S misses. □

Corollary 6.17. *The Noetherian ratio is the ring homomorphism density of the structure sheaf:*

$$\gamma(C) \;=\; \frac{|\{S \subseteq \Lambda : S \textit{ is a ring hom open}\}|}{|\mathrm{Int}(C)|}.$$

Definition 6.18 (Irregularity)**.** The *irregularity* of a finite poset is $\kappa(C) = 1 - 1/\gamma(C)$.

The irregularity measures the density of non-interval ring homomorphism opens: $\kappa = 0$ ($\gamma = 1$) when every ring hom open is an interval (total orders), and $\kappa \to 1$ ($\gamma \to \infty$) when the ring hom opens far exceed the intervals (antichains, random orders).

Remark 6.19 (The convex quotient). Non-isomorphic posets can share identical sets of convex subsets. For example, $\{0 < 1 < 2,\ 3 \text{ isolated}\}$ and $\{0 < 1 < 2,\ 0 < 3\}$ have the same 14 convex subsets but different order relations. The map from posets to convex structures is many-to-one: it extracts the geometric content while discarding combinatorial detail. This is the analogue of the fact in differential geometry that distinct point-sets can underlie the same metric. The sheaf determines a *convex equivalence class*, and γ is an invariant of this class, not of the poset.

This identity unifies the three threads of the paper:

- *Combinatorics*: γ counts the causally convex subsets.
- *Algebra*: the bridge theorem proves these are exactly the ring homomorphism opens.
- *Geometry*: the width scaling law shows their count exhibits a dichotomy between bounded-width and linear-width posets.

The complete reduction is a sequence of proved arrows:

$$\text{Poset} \xrightarrow{\text{many-to-one}} \text{Convex structure} \overset{\text{Thm. 6.16}}{=\!=} \text{Sheaf structure} \xrightarrow{\gamma} \text{Dimension.}$$

The first arrow is a quotient that extracts geometric content. The equality is the bridge theorem. The last arrow is the width scaling law.

A poset with $\gamma \approx 1$ (equivalently $\kappa \approx 0$) has a *uniform sheaf*: the ring homomorphism opens are generated by intervals, and the algebraic structure is simple. A poset with large γ (equivalently $\kappa \to 1$) has a *complex sheaf*: many non-interval ring homomorphism opens create additional local algebraic structure.

7 Non-Abelian Gauge Connection and Holonomy

The vanishing of abelian Čech H^1 for the Benincasa–Dowker curvature presheaf [2] indicates that discrete curvature is fundamentally a connection phenomenon. We construct a non-abelian gauge connection on CSpec whose holonomy detects curvature directly.

7.1 Transition matrices

Given a spatial-wedge cover $\{U_s\}$ of a discretized spacetime, the interval-count matrix A_s with entries $(A_s)_{x,k} = N_k(x, U_s)$—counting $(k{+}2)$-element causal intervals above x within U_s—provides a local frame for the causal geometry. On overlaps $U_s \cap U_{s+1}$, the *transition matrix*

$$T_{s,\,s+1} = A_s^+\, A_{s+1}$$

transforms the interval description from frame s to frame $s+1$, where A_s^+ denotes the Moore–Penrose pseudoinverse restricted to the overlap.

Theorem 7.1 (Functoriality)**.** *The transition matrices satisfy the composition law*

$$T_{p,q} \cdot T_{q,r} \;=\; \mathbf{e}_q$$

for consecutive overlapping wedges p, q, r, where $\mathbf{e}_q$ is the junction idempotent at wedge q. This is the defining property of a discrete connection: parallel transport through q composes associatively, and the junction idempotent acts as the identity on the fiber at q. The interval-projection assignment is therefore functorial *from the path category of the cover to the matrix algebra. All 12 component theorems (associativity, idempotency, composition, naturality, and the resulting functoriality) are proved in Lean 4 [12] with zero* `sorry`*.*

This addresses the concern that the pseudoinverse construction is "ad hoc": it is not merely a matrix computation but a functor satisfying the same composition law as parallel transport in differential geometry.

7.2 The Wilson loop

Definition 7.2. For a closed loop $\gamma = (s_0, s_1, \ldots, s_n = s_0)$ in the cover, the *holonomy* is $\mathrm{Hol}(\gamma) = \prod_i T_{s_i,s_{i+1}}$ and the *Wilson loop* is

$$W_\gamma = \frac{\mathrm{tr}(\mathrm{Hol}(\gamma))}{\dim(\mathrm{Hol}(\gamma))}.$$

The Wilson loop does not depend on the choice of starting wedge (gauge invariance), as follows from cyclicity of the trace. A general proof requires showing the pseudoinverse construction is well-defined; we verify gauge invariance computationally for all tested configurations.

Theorem 7.3 (Wilson Loop Trace)**.** *On discretized cylinders $[c] \times [t]$ with spatial circumference c and temporal depth t, the holonomy trace of the interval-projection connection satisfies*

$$\mathrm{tr}(\mathrm{Hol}) \;=\; t - 1,$$

independent of the spatial circumference c. The independence from spatial circumference follows from the product structure of the interval $[(0,0),(c{-}1,t{-}2)]$.

Proof. The interval $[(0,0),(c{-}1,t{-}2)]$ in $[c] \times [t]$ has exactly $c \cdot (t{-}1)$ elements; the trace identity follows from the product structure of the transition matrices. Both steps are formally verified [12]. □

The pseudoinverse connection of §7 is a richer construction than the interval-projection version; the following table records its Wilson loop values on specific configurations:

Spatial circumference	Temporal depth	W	tr(Hol)
4	3	2/5	2
5	3	2/5	2
6	4	3/5	3
8	4	3/5	3
10	3	2/5	2

The normalized Wilson loop $W = \mathrm{tr}(\mathrm{Hol})/\dim(\mathrm{Hol})$ divides by the matrix dimension $\dim(\mathrm{Hol}) = 5$ (the number of interval types in the pseudoinverse encoding). The key physical content — independence from spatial circumference, with $\mathrm{tr}(\mathrm{Hol}) = t - 1$ — is shared by both connections and is proved for the interval-projection version by Theorem 7.3.

The independence from spatial circumference is physically correct: on a flat cylinder, the spatial circle carries no intrinsic curvature; the holonomy measures the parallel transport cost of the temporal extent. This is a discrete analogue of the fact that Wilson loop expectation values in gauge theory are area-independent for flat connections. The eigenvalue spectrum of Hol exhibits one eigenvalue near unity—the flat direction, corresponding to the translation-invariant count N_0 of direct causal links—with remaining eigenvalues decaying toward zero. Extending the proof to the full pseudoinverse connection on general (non-cylindrical) spacetimes is left for future work. We note, however, that the physical derivation of the Bekenstein–Hawking area law in [27] depends only on the dimension law and the cylindrical factorization, not on any holonomy computation; the Wilson loop theorem is an independent structural result.

The holonomy carries non-trivial structural information beyond classical invariants. The Y-shape ($0 \le 1$, $0 \le 2$, $0 \le 3$) and the Fork ($0 \le 1$, $0 \le 2$, $1 \le 3$) both have 4 elements, 3 links, and chain length 3, but holonomy weights 16 and 13 respectively (proved in Lean 4 [12]). No classical partial-order invariant (element count, edge count, width, chain length) distinguishes them; the holonomy does.

7.3 Curvature discrimination

The Čech H^2 of the Benincasa–Dowker curvature presheaf discriminates spatial topology where H^1 cannot:

Configuration	$\dim H^0$	$\dim H^1$	$\dim H^2$
Cylinder 5×3	15	0	40
Strip 5×3	15	0	20
Cylinder 6×4	24	0	144
Strip 6×4	24	0	88

The gap $\Delta H^2 = 20$ at scale 5×3 and $\Delta H^2 = 56$ at scale 6×4 grows with the size of the topological feature.

8 The Arithmetic Bridge

The divisibility lattice $(\mathbb{Z}^+ \cap [1, N], |)$, for any fixed N, is a finite partial order and therefore a causal algebra in the sense of Definition 2.1.

Theorem 8.1 (Arithmetic Bridge). *For any $N \geq 1$, the truncated divisibility lattice $(\mathbb{Z}^+ \cap [1, N], |)$ satisfies:*

1. *It is a causal algebra under the divisibility order.*
2. *$\mu * \zeta = 1$ in the incidence algebra, where μ is the number-theoretic Möbius function.*
3. *$\mu(p) = -1$ for every prime $p \leq N$.*
4. *Every prime $p \leq N$ generates a causally prime upset in* CSpec*, corresponding to the prime ideals of* $\mathrm{Spec}(\mathbb{Z})$.

Proposition 8.2 (CSpec is strictly richer than Spec). *For the divisibility lattice* $\mathrm{div}(12) = (\{1, 2, 3, 4, 6, 12\}, |)$, $|\mathrm{CSpec}| = 8$ *while* $\mathrm{Spec}(\mathbb{Z}/12\mathbb{Z})$ *has* 2 *prime ideals (corresponding to primes* 2 *and* 3*). The prime upsets embed into* CSpec*, but* CSpec *also contains "composite upsets" — causally convex upsets generated by composite elements — with no* Spec *analogue. Proved in Lean 4 [12].*

This shows that the $\sim$ notation in the comparison table (Section 10) is precisely correct: CSpec contains Spec as a subset of its points but is strictly larger. The additional points encode divisibility structure invisible to the ring-theoretic spectrum.

Theorem 8.3 (CC Doubling). *On the truncated divisibility lattice $(\mathbb{Z}^+ \cap [1, N], |)$, for every prime $p \leq N$:*

$$|\mathrm{CC}([1, p])| \;=\; 2 \cdot |\mathrm{CC}([1, p-1])|.$$

For composite N, the ratio $|\mathrm{CC}([1, N])|/|\mathrm{CC}([1, N-1])| > 1$ but is strictly less than 2.

Proof. Let p be prime. Its only divisors in $[1, p]$ are 1 and p itself. For any convex $S \subseteq [1, p-1]$, the set $S \cup \{p\}$ is convex in $[1, p]$: if $a, b \in S \cup \{p\}$ and $a \mid c \mid b$ with $c \in [1, p]$, then either $b < p$ (reducing to convexity of S) or $b = p$, in which case $c \mid p$ forces $c \in \{1, p\} \subseteq S \cup \{p\}$. Conversely, removing p from a convex set preserves convexity. So $\mathrm{CC}([1, p])$ bijects with $\mathrm{CC}([1, p-1]) \times \{0, 1\}$. □

Corollary 8.4. $\log_2 |\mathrm{CC}([1, N])| = \pi(N) + \log_2 C(N)$*, where π is the prime counting function and $C(N) \geq 1$ accounts for composite corrections. The Noetherian ratio of the divisibility lattice encodes $\pi(N)$.*

9 Formal Verification

All theorems in this paper (except where noted as computational) are formally verified using the Lean 4 proof assistant with Mathlib [11]. The verification covers the full proof chain from Dilworth's theorem (via Hall's marriage theorem) through the width scaling law, the sheaf-γ bridge, the universal property of CSpec, and the Wilson loop trace theorem for the interval-projection connection. The pseudoinverse connection results (Čech H^2 discrimination, eigenvalue spectra, W values) are computational. Source code is available at [12].

10 Discussion and Open Problems

10.1 Correspondence with scheme theory

The causal-algebraic framework adapts several elements of scheme theory to partial orders, with the following correspondences (proved $\longleftrightarrow$; analogical $\sim$):

Commutative Algebra		Causal-Algebraic Geometry
Commutative ring R	$\longleftrightarrow$	Causal algebra $(A, \{\mathbf{e}_\alpha\}, \leq)$
$\mathrm{Spec}(R)$	$\sim$	$\mathrm{CSpec}(A)$
Localization $R_\mathfrak{p}$	$\sim$	Corner algebra $\mathbf{e}_S A\, \mathbf{e}_S$
Structure sheaf $\mathcal{O}$	$\sim$	Corner algebra sheaf
Sheaf cohomology	$\sim$	Order-complex cohomology
$\mathrm{Spec}(\mathbb{Z})$	$\sim$	$\mathrm{CSpec}(k[\mathrm{Div}(N)])$ (finite truncation)
Noetherian condition	$\sim$	$\gamma \approx 1$
Spectrum determined by ring	$\longleftrightarrow$	Spectrum is terminal object (Thm. 3.9)

Entries marked $\longleftrightarrow$ are proved correspondences; entries marked $\sim$ are structural analogies that motivate the construction but are not claimed as formal equivalences. In particular, corner algebras play a role analogous to localizations but are not shown to satisfy the same universal property.

10.2 Main theorem

The results of this paper combine into the following single statement, which we record for ease of citation.

Theorem 10.1 (Main Theorem)**.** *Let $(C, \leq)$ be a finite poset of width w on n elements.*

1. (Universal property.) $\mathrm{CSpec}(k[C])$ *is the terminal object in the category of causal topologies on $k[C]$: every compatible topology maps to* CSpec*, the morphism is unique, and no strictly larger compatible topology exists.*
2. (Bridge.) *A subset $S \subseteq C$ is a ring homomorphism open of the structure sheaf if and only if S is causally convex.*
3. (Scaling.) $2^w \leq |\mathrm{CC}(C)| \leq (\lceil n/w \rceil^2 + 1)^w$. *Since $|\mathrm{Int}(C)| \geq n^2/(2w)$ for any poset of width w on n elements (Cauchy-Schwarz; proved in Lean 4 [12]), this gives $\gamma = O(n^{2(w-1)})$ for fixed w. A universal lower bound $\gamma \geq 2^w/n^2$ follows from $|\mathrm{CC}| \geq 2^w$ and $|\mathrm{Int}| \leq n^2$.*
4. (Gap.) *If $w = O(1)$, then γ is polynomial in n. If $w = \Omega(n)$, then γ is exponential in n.*

The Noetherian ratio $\gamma = |\mathrm{CC}(C)|/|\mathrm{Int}(C)|$ is simultaneously a combinatorial invariant (counting convex subsets), a sheaf invariant (counting ring homomorphism opens), and a width detector (polynomial degree determined by width). The irregularity $\kappa = 1 - 1/\gamma$ measures the fraction of non-interval ring homomorphism opens. All parts are unconditional.

10.3 Open problems

1. **Dimensional scaling.** The dimension law $\log|\mathrm{CC}([m]^d)| = \Theta(m^{d-1})$ is proved for all $d \geq 2$ in [26] via a tiling inequality and an ideal/filter injection. The width scaling law gives $\gamma = O(n^{2(w-1)})$ for fixed w; this bound is sharp on tested examples. For the d-dimensional grid $[m]^d$ on $n = m^d$ elements, the width is $w = m^{d-1} = n^{(d-1)/d}$. Note that the $O(n^{2(w-1)})$ bound was derived for fixed w; substituting $w \sim n^{(d-1)/d}$ gives a heuristic, not a theorem. The rigorous result for grids is the dimension law $\log|\mathrm{CC}([m]^d)| = \Theta(m^{d-1})$, proved directly. Computing exact growth constants ρ_d for $d \geq 3$ and exact formulas for $\gamma([m]^d)$ remain open. Combined with the upper bound $\log|\mathrm{CC}| \leq w \cdot \log(\lceil n/w \rceil^2 + 1)$ from Theorem 1.8, the dimension law yields a uniqueness result: the entropy exponent α in $\log|\mathrm{CC}| = \Theta(n^\alpha)$ determines the dimension via $d = 1/(1-\alpha)$. In particular, the Bekenstein–Hawking area law $S \sim m^{d-2}$ for equilibrium configurations implies $d = 4$; see [27] for details.

2. **Continuum limit via poset limits.** Janson [20] and Hladký, Máthé, Patel, and Pikhurko [21] developed the theory of poset limits ("posons"), in which a sequence of finite posets converges when all finite substructure densities converge. The natural formulation of the continuum limit for CSpec is: for a sequence of Poisson processes at increasing density in a Lorentzian manifold, γ should be expressible as a functional of the limiting poson, and the CSpec topology should converge to a pro-CSpec analogous to Grothendieck's pro-étale constructions. Brightwell and Luczak [13] note that Bombelli's conjecture [22]—that substructure densities distinguish manifold regions up to measure-preserving diffeomorphism—is the poset-limit formulation of the Hauptvermutung.

3. **Contiguity for $d > 2$.** For $d = 2$, the uniform measure on 2-dimensional orders is contiguous with a Poisson process in $\mathbb{M}^2$ [23, 16], providing the only existing proof that a purely combinatorial measure produces manifold-like structure without referencing any continuous geometry. For $d > 2$, the analogous contiguity is open [13]. If the gap theorem could be strengthened to show that the *uniform measure on posets with polynomial* γ concentrates on manifold-like orders, this would be progress on this open problem.

4. **Gibbs measures and discrete gravitational dynamics.** Brightwell and Luczak [19] showed that order-invariant measures on causal sets satisfy equations analogous to the DLR equations of statistical mechanics (Theorem 4.2 of [13]). If one defines a "causal Hamiltonian" $H(C) = -\sum_\alpha \kappa_{\mathrm{BD}}(\alpha)$ using the Benincasa–Dowker curvature, the Gibbs measure $e^{-\beta H}$ on the space of finite causal sets is a discrete path integral for gravity. Extremizing this measure should select posets that extremize total curvature—i.e., solutions to the discrete Einstein equations. The DLR framework provides the rigorous language; the connection to CSpec and γ would come from expressing the curvature functional in terms of the structure sheaf.

5. **Causal Grothendieck site.** Replace the Alexandrov topology with a Grothendieck topology whose covering sieves are families of causal embeddings, potentially yielding non-trivial abelian H^1.

6. **Higher-dimensional scaling.** Compute γ for Poisson sprinklings in d-dimensional Minkowski space for $d = 2, 3, 4$ and verify the predicted scaling.

7. **Join-irreducible generalization.** The CC Doubling Theorem generalizes: for any finite lattice and any join-irreducible element j, adding j doubles $|\mathrm{CC}|$. This gives a structural decomposition $|\mathrm{CC}(L)| = 2^{j(L)} \cdot C(L)$ where $j(L)$ is the number of join-irreducibles and $C(L)$ accounts for non-join-irreducible corrections, with applications to Boolean lattices, partition lattices, and face lattices of polytopes.

Acknowledgments

The formal verification builds on the Mathlib library [11].

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